

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415659
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Unger; Jens Holger

Lid from paper, paperboard or paper-like material, and method for producing a lid

Abstract

A lid from paper, paperboard or paper-like material for a container, wherein the lid is produced from an integral planar blank and has a lid plate and a lid collar. The lid collar proceeds from a periphery of the lid plate, and the lid collar is formed from at least two plies of the planar blank. A first ply of the lid collar, proceeding from the periphery of the lid plate, extends to a periphery of the lid collar, and wherein a second ply of the lid collar, from the periphery of the lid collar, on a radially inner side of the lid collar extends to a periphery of the second ply of the lid collar. The periphery of the second ply of the lid collar at least in portions is received in a groove in the region of the periphery of the lid plate.

Inventors: Unger; Jens Holger (Esslingen, DE)

Applicant: Michael Hörauf Maschinenfabrik GmbH und Co. KG (Donzdorf, DE)

Family ID: 1000008818553

Assignee: MICHAEL HÖRAUF MASCHINENFABRIK GMBH UND CO. KG (Donzdorf, DE)

Appl. No.: 18/172552

Filed: February 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230264866 A1	Aug. 24, 2023

Foreign Application Priority Data

DE	102022201836.2	Feb. 22, 2022
EP	23156018	Feb. 10, 2023

Publication Classification

Int. Cl.: **B65D43/02** (20060101); **B31B50/28** (20170101); **B31B50/62** (20170101); **B65D43/06** (20060101); B31B120/10 (20170101); B31D1/00 (20170101); B31D5/00 (20170101); B65D41/02 (20060101)

U.S. Cl.:

CPC **B65D43/0222** (20130101); **B31B50/28** (20170801); **B31B50/62** (20170801); **B65D43/065** (20130101); B31B2120/10 (20170801); B65D2543/00092 (20130101); B65D2543/00268 (20130101)

Field of Classification Search

CPC: B31B (2120/10); B31B (50/28); B31B (50/62); B31D (1/005); B31D (5/00); B65D (2251/06); B65D (23/106); B65D (25/32); B65D (25/48); B65D (2543/00092); B65D (2543/00268); B65D (2543/00768); B65D (2581/055); B65D (41/02); B65D (41/48); B65D (43/0212); B65D (43/022); B65D (43/0222); B65D (43/065); B65D (47/20); B65D (47/36); B65D (51/18); B65D (51/247); B65D (55/024); B65D (55/0863); B65D (75/004); B65D (81/058); B65D (1/00); B65D (1/02); B65D (1/0207); B65D (1/0215); B65D (1/0223); B65D (1/023); B65D (1/0238); B65D (1/0246); B65D (1/0292); B65D (1/095); B65D (1/10); B65D (1/12); B65D (1/16); B65D (1/165); B65D (1/22); B65D (1/225); B65D (1/24); B65D (1/26); B65D (1/265); B65D (1/28); B65D (1/30); B65D (1/32); B65D (1/323); B65D (1/326); B65D (1/34); B65D (1/36); B65D (1/40); B65D (1/42); B65D (1/44); B65D (1/46); B65D (1/48); B65D (11/00); B65D (11/02); B65D (11/04); B65D (11/08); B65D (11/10); B65D (11/18); B65D (11/1873); B65D (11/188); B65D (11/20); B65D (13/04); B65D (15/02); B65D (15/04); B65D (15/06); B65D (15/08); B65D (15/10); B65D (15/20); B65D (15/22); B65D (15/24); B65D (17/00); B65D (17/02); B65D (17/04); B65D (17/28); B65D (17/30); B65D (17/34); B65D (17/36); B65D (17/401); B65D (17/4011); B65D (17/4012); B65D (17/4014); B65D (17/402); B65D (17/404); B65D (17/50); B65D (17/501); B65D (17/502); B65D (17/505); B65D (17/506); B65D (17/508); B65D (19/0004); B65D (19/0012); B65D (19/06); B65D (19/18); B65D (19/20); B65D (19/38); B65D (19/42); B65D (21/00); B65D (21/02); B65D (21/0201); B65D (21/0202); B65D (21/0204); B65D (21/0206); B65D (21/0209); B65D (21/0212); B65D (21/0213); B65D (21/0215); B65D (21/0217); B65D (21/0219); B65D (21/022); B65D (21/0222); B65D (21/0223); B65D (21/023); B65D (21/0231); B65D (21/0233); B65D (21/0234); B65D (21/0235); B65D (21/0238); B65D (21/04); B65D (21/064); B65D (21/083); B65D (21/086); B65D (2201/00); B65D (2203/00); B65D (2203/02); B65D (2203/06); B65D (2203/10); B65D (2203/12); B65D (2205/00); B65D (2205/02); B65D (2205/025); B65D (2207/00); B65D (2209/00); B65D (2211/00); B65D (2213/00); B65D (2215/00); B65D (2215/02); B65D (2215/04); B65D (2215/06); B65D (2221/00); B65D (2231/00); B65D (2231/002); B65D (2231/004); B65D (2231/005); B65D (2231/007); B65D (2231/02); B65D (2231/022); B65D (2231/025); B65D (2251/0003); B65D (2251/0012); B65D (2251/0015); B65D (2251/0018); B65D (2251/0021); B65D (2251/0025); B65D (2251/0028); B65D (2251/0031); B65D (2251/0046); B65D (2251/005); B65D (2251/0056); B65D (2251/0059); B65D (2251/0068); B65D (2251/0071); B65D (2251/0078); B65D (2251/0081); B65D (2251/0087); B65D (2251/009); B65D (2251/0093); B65D (2251/0096); B65D (2251/01); B65D (2251/023); B65D (2251/04); B65D (2251/08); B65D (2251/1008); B65D (2251/1016); B65D (2251/1025); B65D (2251/1033); B65D

(2251/1041); B65D (2251/105); B65D (2251/1058); B65D (2251/1066); B65D (2251/1083); B65D (2251/20); B65D (2251/205); B65D (2255/00); B65D (2255/20); B65D (2275/00); B65D (23/00); B65D (23/001); B65D (23/02); B65D (23/04); B65D (23/085); B65D (23/0878); B65D (23/10); B65D (23/102); B65D (23/12); B65D (23/14); B65D (2301/10); B65D (2301/20); B65D (2313/00); B65D (2313/02); B65D (2313/04); B65D (2313/10); B65D (2401/00); B65D (2401/05); B65D (2401/10); B65D (2401/15); B65D (2401/20); B65D (2401/25); B65D (2401/30); B65D (2401/35); B65D (2401/45); B65D (2401/50); B65D (2401/60); B65D (25/00); B65D (25/005); B65D (25/02); B65D (25/04); B65D (25/06); B65D (25/08); B65D (25/082); B65D (25/10); B65D (25/101); B65D (25/105); B65D (25/107); B65D (25/108); B65D (25/14); B65D (25/16); B65D (25/20); B65D (25/205); B65D (25/22); B65D (25/24); B65D (25/28); B65D (25/2805); B65D (25/282); B65D (25/2835); B65D (25/2841); B65D (25/2852); B65D (25/2858); B65D (25/287); B65D (25/2885); B65D (25/2888); B65D (25/2894); B65D (25/2897); B65D (25/30); B65D (25/34); B65D (25/36); B65D (25/38); B65D (25/40); B65D (25/42); B65D (25/52); B65D (25/54); B65D (25/56); B65D (2517/0008); B65D (2517/0013); B65D (2517/0014); B65D (2517/0025); B65D (2517/0041); B65D (2517/0043); B65D (2517/0044); B65D (2517/0056); B65D (2517/0061); B65D (2517/0062); B65D (2517/0065); B65D (2517/0082); B65D (2517/0083); B65D (2517/0085); B65D (2517/0086); B65D (2517/0089); B65D (2517/0092); B65D (2517/0098); B65D (2517/5002); B65D (2517/5008); B65D (2517/501); B65D (2517/5018); B65D (2517/5027); B65D (2517/5029); B65D (2517/5032); B65D (2517/5037); B65D (2517/5054); B65D (2517/5072); B65D (2517/5083); B65D (2517/5086); B65D (2517/5089); B65D (2517/5097); B65D (2519/00019); B65D (2519/00024); B65D (2519/00029); B65D (19/00034); B65D (2519/00054); B65D (2519/00059); B65D (2519/00064); B65D (2519/00069); B65D (2519/00089); B65D (2519/00104); B65D (2519/00159); B65D (2519/00164); B65D (2519/00174); B65D (2519/00184); B65D (2519/00194); B65D (2519/00199); B65D (2519/00208); B65D (2519/00253); B65D (2519/00268); B65D (2519/00273); B65D (2519/00278); B65D (2519/00288); B65D (2519/00293); B65D (2519/00303); B65D (2519/00318); B65D (2519/00333); B65D (2519/00338); B65D (2519/00343); B65D (2519/00348); B65D (2519/00407); B65D (2519/00412); B65D (2519/00427); B65D (2519/00452); B65D (2519/00497); B65D (2519/00502); B65D (2519/00547); B65D (2519/00552); B65D (2519/00557); B65D (2519/00562); B65D (2519/00567); B65D (2519/00572); B65D (2519/00582); B65D (2519/00587); B65D (2519/00592); B65D (2519/00597); B65D (2519/00611); B65D (2519/00616); B65D (2519/00621); B65D (2519/00641); B65D (2519/00656); B65D (2519/00661); B65D (2519/00706); B65D (2519/00711); B65D (2519/00726); B65D (2519/00731); B65D (2519/00786); B65D (2519/008); B65D (2519/00805); B65D (2519/0081); B65D (2519/00855); B65D (2519/0086); B65D (2519/00875); B65D (2519/0091); B65D (2519/0093); B65D (2519/0096); B65D (2519/00965); B65D (2519/00975); B65D (2519/0098); B65D (2519/0099); B65D (2519/00995); B65D (2525/281); B65D (2543/00018); B65D (2543/00027); B65D (2543/00037); B65D (2543/00046); B65D (2543/00055); B65D (2543/00064); B65D (2543/00074); B65D (2543/00083); B65D (2543/00101); B65D (2543/0012); B65D (2543/00148); B65D (2543/00157); B65D (2543/00175); B65D (2543/00185); B65D (2543/00194); B65D (2543/00203); B65D (2543/00212); B65D (2543/00222); B65D (2543/00231); B65D (2543/0024); B65D (2543/0025); B65D (2543/00277); B65D (2543/00287); B65D (2543/00296); B65D (2543/00305); B65D (2543/00314); B65D (2543/00324); B65D (2543/00333); B65D (2543/00342); B65D (2543/00351); B65D (2543/00361); B65D (2543/0037); B65D (2543/00379); B65D (2543/00388); B65D (2543/00398); B65D (2543/00407); B65D (2543/00416); B65D (2543/00425); B65D

(2543/00435); B65D (2543/00453); B65D (2543/00462); B65D (2543/00472); B65D (2543/00481); B65D (2543/0049); B65D (2543/005); B65D (2543/00509); B65D (2543/00518); B65D (2543/00527); B65D (2543/00537); B65D (2543/00546); B65D (2543/00555); B65D (2543/00564); B65D (2543/00574); B65D (2543/00583); B65D (2543/00601); B65D (2543/0062); B65D (2543/00629); B65D (2543/00638); B65D (2543/00648); B65D (2543/00657); B65D (2543/00666); B65D (2543/00675); B65D (2543/00685); B65D (2543/00694); B65D (2543/00712); B65D (2543/00731); B65D (2543/0074); B65D (2543/0075); B65D (2543/00759); B65D (2543/00787); B65D (2543/00796); B65D (2543/00805); B65D (2543/00814); B65D (2543/00824); B65D (2543/00833); B65D (2543/00842); B65D (2543/00851); B65D (2543/00879); B65D (2543/00888); B65D (2543/00898); B65D (2543/00907); B65D (2543/00916); B65D (2543/00925); B65D (2543/00944); B65D (2543/00953); B65D (2543/00962); B65D (2543/00972); B65D (2543/00981); B65D (2543/0099); B65D (2547/063); B65D (2565/385); B65D (2565/388); B65D (2571/00037); B65D (2571/00049); B65D (2571/00055); B65D (2571/00074); B65D (2571/00111); B65D (2571/00141); B65D (2571/00154); B65D (2571/00222); B65D (2571/0024); B65D (2571/00277); B65D (2571/0029); B65D (2571/00327); B65D (2571/00339); B65D (2571/00388); B65D (2571/00444); B65D (2571/0045); B65D (2571/00487); B65D (2571/00524); B65D (2571/0066); B65D (2571/00666); B65D (2571/00672); B65D (2571/00679); B65D (2571/00728); B65D (2571/0079); B65D (2571/00796); B65D (2571/00802); B65D (2571/00839); B65D (2571/00882); B65D (2571/00901); B65D (2571/00919); B65D (2571/00932); B65D (2571/00956); B65D (2575/3227); B65D (2575/3245); B65D (2573/3263); B65D (2575/583); B65D (2575/586); B65D (2577/047); B65D (2577/2008); B65D (2577/2016); B65D (2577/2025); B65D (2577/2041); B65D (2577/205); B65D (2577/2058); B65D (2577/2066); B65D (2577/2083); B65D (2577/2091); B65D (2581/053); B65D (2581/3404); B65D (2581/3405); B65D (2581/3406); B65D (2581/3408); B65D (2581/3416); B65D (2581/3422); B65D (2581/3428); B65D (2581/3435); B65D (2581/3441); B65D (2581/3443); B65D (2581/345); B65D (2581/3464); B65D (2581/3472); B65D (2581/3477); B65D (2581/3479); B65D (2581/3487); B65D (2581/3494); B65D (2581/3495); B65D (2583/0409); B65D (2583/0481); B65D (2583/0495); B65D (2585/00); B65D (2585/36); B65D (2585/363); B65D (2585/366); B65D (2585/545); B65D (2585/56); B65D (2585/689); B65D (2585/88); B65D (2590/046); B65D (27/04); B65D (27/08); B65D (27/14); B65D (27/16); B65D (27/28); B65D (27/38); B65D (29/00); B65D (29/04); B65D (3/00); B65D (3/02); B65D (3/04); B65D (3/06); B65D (3/08); B65D (3/10); B65D (3/12); B65D (3/14); B65D (3/16); B65D (3/18); B65D (3/20); B65D (3/22); B65D (3/24); B65D (3/26); B65D (3/262); B65D (3/263); B65D (3/268); B65D (3/28); B65D (3/30); B65D (31/00); B65D (31/02); B65D (31/10); B65D (31/12); B65D (33/00); B65D (33/001); B65D (33/06); B65D (33/14); B65D (33/16); B65D (33/1675); B65D (33/18); B65D (33/20); B65D (33/2508); B65D (33/30); B65D (35/02); B65D (35/10); B65D (35/12); B65D (35/14); B65D (35/242); B65D (35/36); B65D (35/44); B65D (39/0005); B65D (39/0011); B65D (39/0052); B65D (39/0058); B65D (39/007); B65D (39/0076); B65D (39/02); B65D (39/025); B65D (39/04); B65D (39/08); B65D (39/10); B65D (39/14); B65D (39/16); B65D (41/00); B65D (41/005); B65D (41/023); B65D (41/026); B65D (41/04); B65D (41/0407); B65D (41/0414); B65D (41/0421); B65D (41/0428); B65D (41/0435); B65D (41/0442); B65D (41/045); B65D (41/0457); B65D (41/0464); B65D (41/0471); B65D (41/0478); B65D (41/0485); B65D (41/0492); B65D (41/06); B65D (41/065); B65D (41/08); B65D (41/12); B65D (41/125); B65D (41/16); B65D (41/165); B65D (41/17); B65D (41/18); B65D (41/185); B65D (41/20); B65D (41/22); B65D (41/26); B65D (41/28); B65D (41/32); B65D (41/325); B65D (41/34); B65D

(41/3404); B65D (41/3414); B65D (41/3419); B65D (41/3423); B65D (41/3428); B65D (41/3433); B65D (41/3438); B65D (41/3442); B65D (41/3447); B65D (41/3457); B65D (41/348); B65D (41/42); B65D (41/44); B65D (41/46); B65D (41/465); B65D (41/485); B65D (41/56); B65D (41/58); B65D (41/62); B65D (43/00); B65D (43/02); B65D (43/0202); B65D (43/0204); B65D (43/0206); B65D (43/0208); B65D (43/021); B65D (43/0214); B65D (43/0216); B65D (43/0218); B65D (43/0225); B65D (43/0227); B65D (43/0229); B65D (43/0231); B65D (43/0233); B65D (43/0235); B65D (43/0237); B65D (43/0249); B65D (43/0254); B65D (43/0256); B65D (43/0258); B65D (43/0262); B65D (43/0264); B65D (43/0268); B65D (43/027); B65D (43/0272); B65D (43/0285); B65D (43/0297); B65D (43/06); B65D (43/08); B65D (43/12); B65D (43/14); B65D (43/16); B65D (43/161); B65D (43/162); B65D (43/163); B65D (43/164); B65D (43/165); B65D (43/166); B65D (43/168); B65D (43/169); B65D (43/18); B65D (43/20); B65D (43/22); B65D (43/24); B65D (43/26); B65D (43/262); B65D (43/265); B65D (43/267); B65D (45/00); B65D (45/02); B65D (45/025); B65D (45/16); B65D (45/18); B65D (45/20); B65D (45/22); B65D (45/24); B65D (45/30); B65D (45/32); B65D (45/322); B65D (45/325); B65D (45/34); B65D (45/345); B65D (47/00); B65D (47/02); B65D (47/04); B65D (47/043); B65D (47/06); B65D (47/061); B65D (47/063); B65D (47/065); B65D (47/066); B65D (47/08); B65D (47/0804); B65D (47/0809); B65D (47/0814); B65D (47/0823); B65D (47/0828); B65D (47/0833); B65D (47/0838); B65D (47/0842); B65D (47/0847); B65D (47/0852); B65D (47/0857); B65D (47/0866); B65D (47/0871); B65D (47/0885); B65D (47/089); B65D (47/0895); B65D (47/10); B65D (47/103); B65D (47/12); B65D (47/121); B65D (47/122); B65D (47/123); B65D (47/128); B65D (47/14); B65D (47/147); B65D (47/148); B65D (47/18); B65D (47/2006); B65D (47/2012); B65D (47/2018); B65D (47/2031); B65D (47/2062); B65D (47/2075); B65D (47/2081); B65D (47/2087); B65D (47/242); B65D (47/243); B65D (47/245); B65D (47/247); B65D (47/248); B65D (47/249); B65D (47/26); B65D (47/261); B65D (47/263); B65D (47/265); B65D (47/28); B65D (47/283); B65D (47/286); B65D (47/305); B65D (47/32); B65D (47/40); B65D (47/42); B65D (47/44); B65D (49/02); B65D (49/04); B65D (49/06); B65D (49/08); B65D (49/12); B65D (5/00); B65D (5/0005); B65D (5/001); B65D (5/0015); B65D (5/0025); B65D (5/003); B65D (5/0035); B65D (5/0085); B65D (5/02); B65D (5/0209); B65D (5/0218); B65D (5/0227); B65D (5/0254); B65D (5/0263); B65D (5/0272); B65D (5/0281); B65D (5/029); B65D (5/06); B65D (5/062); B65D (5/065); B65D (5/067); B65D (5/068); B65D (5/08); B65D (5/10); B65D (5/12); B65D (5/14); B65D (5/16); B65D (5/18); B65D (5/20); B65D (5/2004); B65D (5/2009); B65D (5/2019); B65D (5/2023); B65D (5/2033); B65D (5/2038); B65D (5/2042); B65D (5/2047); B65D (5/2052); B65D (5/2057); B65D (5/2066); B65D (5/2076); B65D (5/209); B65D (5/2095); B65D (5/22); B65D (5/24); B65D (5/241); B65D (5/242); B65D (5/244); B65D (5/246); B65D (5/247); B65D (5/248); B65D (5/28); B65D (5/30); B65D (5/302); B65D (5/32); B65D (5/321); B65D (5/323); B65D (5/324); B65D (5/326); B65D (5/36); B65D (5/3621); B65D (5/3628); B65D (5/3642); B65D (5/3657); B65D (5/3678); B65D (5/38); B65D (5/40); B65D (5/42); B65D (5/4204); B65D (5/4208); B65D (5/4212); B65D (5/4216); B65D (5/4225); B65D (5/4229); B65D (5/4233); B65D (5/425); B65D (5/4258); B65D (5/4266); B65D (5/4279); B65D (5/4283); B65D (5/4287); B65D (5/4291); B65D (5/4295); B65D (5/44); B65D (5/441); B65D (5/443); B65D (5/445); B65D (5/448); B65D (5/46); B65D (5/46008); B65D (5/46024); B65D (5/4604); B65D (5/46048); B65D (5/46072); B65D (5/4608); B65D (5/46088); B65D (5/46112); B65D (5/46144); B65D (5/46184); B65D (5/48); B65D (5/48002); B65D (5/4802); B65D (5/48022); B65D (5/48024); B65D (5/48026); B65D (5/48034); B65D (5/48038); B65D (5/48042); B65D (5/48044); B65D (5/48048); B65D (5/50); B65D (5/5004); B65D (5/5007); B65D (5/5016); B65D

(5/5019); B65D (5/5021); B65D (5/5028); B65D (5/503); B65D (5/5035); B65D (5/5038); B65D (5/504); B65D (5/5052); B65D (5/5059); B65D (5/5076); B65D (5/52); B65D (5/5226); B65D (5/5253); B65D (5/528); B65D (5/54); B65D (5/5405); B65D (5/541); B65D (5/542); B65D (5/5425); B65D (5/543); B65D (5/5435); B65D (5/544); B65D (5/5445); B65D (5/545); B65D (5/5455); B65D (5/546); B65D (5/548); B65D (5/5485); B65D (5/56); B65D (5/563); B65D (5/566); B65D (5/58); B65D (5/60); B65D (5/62); B65D (5/64); B65D (5/643); B65D (5/66); B65D (5/6602); B65D (5/6605); B65D (5/6608); B65D (5/6611); B65D (5/6614); B65D (5/662); B65D (5/6623); B65D (5/6626); B65D (5/6629); B65D (5/6635); B65D (5/6638); B65D (5/665); B65D (5/6652); B65D (5/6655); B65D (5/6661); B65D (5/6664); B65D (5/6667); B65D (5/6682); B65D (5/6685); B65D (5/6688); B65D (5/6691); B65D (5/6697); B65D (5/68); B65D (5/685); B65D (5/70); B65D (5/701); B65D (5/703); B65D (5/705); B65D (5/706); B65D (5/708); B65D (5/72); B65D (5/721); B65D (5/722); B65D (5/723); B65D (5/724); B65D (5/725); B65D (5/727); B65D (5/728); B65D (5/74); B65D (5/742); B65D (5/744); B65D (5/746); B65D (5/748); B65D (5/749); B65D (50/00); B65D (50/02); B65D (50/04); B65D (50/041); B65D (50/043); B65D (50/045); B65D (50/046); B65D (50/06); B65D (50/061); B65D (50/066); B65D (50/067); B65D (50/068); B65D (51/00); B65D (51/002); B65D (51/005); B65D (51/007); B65D (51/04); B65D (51/08); B65D (51/12); B65D (51/14); B65D (51/145); B65D (51/16); B65D (51/611); B65D (51/1616); B65D (51/1622); B65D (51/1627); B65D (51/1638); B65D (51/1644); B65D (51/165); B65D (51/1661); B65D (51/1666); B65D (51/1672); B65D (51/1683); B65D (51/1688); B65D (51/1694); B65D (51/185); B65D (51/20); B65D (51/22); B65D (51/221); B65D (51/222); B65D (51/223); B65D (51/224); B65D (51/225); B65D (51/227); B65D (51/228); B65D (51/24); B65D (51/241); B65D (51/242); B65D (51/243); B65D (51/245); B65D (51/246); B65D (51/248); B65D (51/249); B65D (51/26); B65D (51/28); B65D (51/2807); B65D (51/2814); B65D (51/2821); B65D (51/2828); B65D (51/2835); B65D (51/285); B65D (51/2857); B65D (51/2864); B65D (51/2885); B65D (51/2892); B65D (51/30); B65D (51/32); B65D (53/00); B65D (53/02); B65D (53/04); B65D (53/06); B65D (53/08); B65D (55/00); B65D (55/02); B65D (55/022); B65D (55/026); B65D (55/04); B65D (55/06); B65D (55/063); B65D (55/066); B65D (55/08); B65D (55/0818); B65D (55/0827); B65D (55/0845); B65D (55/0854); B65D (55/089); B65D (55/10); B65D (55/12); B65D (55/14); B65D (55/16); B65D (57/003); B65D (59/02); B65D (59/04); B65D (59/06); B65D (59/08); B65D (61/00); B65D (63/10); B65D (63/1009); B65D (65/00); B65D (65/02); B65D (65/06); B65D (65/10); B65D (65/14); B65D (65/18); B65D (65/22); B65D (65/24); B65D (65/38); B65D (65/40); B65D (65/403); B65D (65/42); B65D (65/44); B65D (65/46); B65D (65/463); B65D (65/466); B65D (7/04); B65D (7/06); B65D (7/24); B65D (7/26); B65D (7/30); B65D (7/36); B65D (7/42); B65D (71/00); B65D (71/0022); B65D (71/004); B65D (71/0096); B65D (71/02); B65D (71/04); B65D (71/066); B65D (71/08); B65D (71/10); B65D (71/20); B65D (71/38); B65D (71/40); B65D (71/406); B65D (71/46); B65D (71/70); B65D (73/0028); B65D (73/0057); B65D (73/0064); B65D (73/0071); B65D (73/0078); B65D (75/00); B65D (75/002); B65D (75/008); B65D (75/02); B65D (75/04); B65D (75/06); B65D (75/08); B65D (75/12); B65D (75/14); B65D (75/20); B65D (75/22); B65D (75/26); B65D (75/28); B65D (75/30); B65D (75/305); B65D (75/32); B65D (75/322); B65D (75/323); B65D (75/325); B65D (75/326); B65D (75/327); B65D (75/34); B65D (75/36); B65D (75/366); B65D (75/367); B65D (75/38); B65D (75/40); B65D (75/42); B65D (75/52); B65D (75/527); B65D (75/54); B65D (75/56); B65D (75/58); B65D (75/5805); B65D (75/5827); B65D (75/5833); B65D (75/5838); B65D (75/5855); B65D (75/5861); B65D (75/5866); B65D (75/5877); B65D (75/5883); B65D (75/5894); B65D (75/66); B65D (77/00); B65D (77/003); B65D (77/02); B65D (77/04); B65D (77/0413); B65D

(77/042); B65D (77/0426); B65D (77/0433); B65D (77/0446); B65D (77/0453); B65D (77/046); B65D (77/0466); B65D (77/048); B65D (77/0486); B65D (77/0493); B65D (77/06); B65D (77/062); B65D (77/067); B65D (77/08); B65D (77/10); B65D (77/20); B65D (77/2004); B65D (77/2012); B65D (77/2016); B65D (77/202); B65D (77/2024); B65D (77/2028); B65D (77/2032); B65D (77/2036); B65D (77/204); B65D (77/2044); B65D (77/2048); B65D (77/2056); B65D (77/206); B65D (77/2068); B65D (77/2076); B65D (77/208); B65D (77/2084); B65D (77/2088); B65D (77/2096); B65D (77/22); B65D (77/225); B65D (77/24); B65D (77/245); B65D (77/26); B65D (77/28); B65D (77/283); B65D (77/286); B65D (77/30); B65D (77/32); B65D (77/36); B65D (77/38); B65D (79/00); B65D (79/005); B65D (79/0087); B65D (79/02); B65D (81/00); B65D (81/02); B65D (81/022); B65D (81/025); B65D (81/027); B65D (81/03); B65D (81/05); B65D (81/051); B65D (81/052); B65D (81/055); B65D (81/07); B65D (81/113); B65D (81/127); B65D (81/1275); B65D (81/133); B65D (81/18); B65D (81/20); B65D (81/2007); B65D (81/2015); B65D (81/2023); B65D (81/2038); B65D (81/2076); B65D (81/22); B65D (81/24); B65D (81/245); B65D (82/16); B65D (81/263); B65D (81/264); B65D (81/266); B65D (81/267); B65D (81/268); B65D (81/28); B65D (81/32); B65D (81/3205); B65D (81/3211); B65D (81/3216); B65D (81/3222); B65D (81/3233); B65D (81/3244); B65D (81/3255); B65D (81/3272); B65D (81/3288); B65D (38/3294); B65D (81/34); B65D (81/3415); B65D (81/3423); B65D (81/343); B65D (81/3438); B65D (81/3446); B65D (81/3453); B65D (81/3461); B65D (81/3476); B65D (81/3484); B65D (81/36); B65D (81/361); B65D (81/366); B65D (81/368); B65D (81/38); B65D (81/3802); B65D (81/3811); B65D (81/3813); B65D (81/3816); B65D (81/3818); B65D (81/3823); B65D (81/3825); B65D (81/383); B65D (81/3834); B65D (81/3841); B65D (81/3853); B65D (81/386); B65D (81/3865); B65D (81/3867); B65D (81/3869); B65D (81/3874); B65D (81/3876); B65D (81/389); B65D (83/00); B65D (83/0094); B65D (83/02); B65D (83/04); B65D (83/0409); B65D (83/0418); B65D (83/0454); B65D (83/0463); B65D (83/0472); B65D (83/0481); B65D (83/06); B65D (83/08); B65D (83/0805); B65D (83/0811); B65D (83/0817); B65D (83/0835); B65D (83/0841); B65D (83/0847); B65D (83/0858); B65D (83/087); B65D (83/0876); B65D (83/0882); B65D (83/0888); B65D (83/0894); B65D (83/145); B65D (83/20); B65D (83/206); B65D (83/22); B65D (83/226); B65D (83/228); B65D (83/24); B65D (83/28); B65D (83/303); B65D (83/32); B65D (83/345); B65D (83/40); B65D (83/425); B65D (83/44); B65D (83/46); B65D (83/48); B65D (83/64); B65D (83/646); B65D (83/66); B65D (83/75); B65D (83/7532); B65D (83/7535); B65D (83/756); B65D (83/76); B65D (83/761); B65D (83/764); B65D (83/767); B65D (83/768); B65D (83/771); B65D (83/7713); B65D (85/00); B65D (85/02); B65D (85/04); B65D (85/07); B65D (85/10); B65D (85/1009); B65D (85/1018); B65D (85/1027); B65D (85/1036); B65D (85/1045); B65D (85/1048); B65D (85/10484); B65D (85/1054); B65D (85/1056); B65D (85/10564); B65D (85/10566); B65D (85/10568); B65D (85/1063); B65D (85/1081); B65D (85/109); B65D (85/12); B65D (85/18); B65D (85/20); B65D (85/22); B65D (85/24); B65D (85/30); B65D (85/305); B65D (85/307); B65D (85/32); B65D (85/322); B65D (85/324); B65D (85/325); B65D (85/34); B65D (85/36); B65D (85/38); B65D (85/42); B65D (85/50); B65D (85/52); B65D (85/544); B65D (85/60); B65D (85/62); B65D (85/66); B65D (85/67); B65D (85/671); B65D (85/672); B65D (85/68); B65D (85/70); B65D (85/72); B65D (85/73); B65D (85/74); B65D (85/76); B65D (85/78); B65D (85/80); B65D (85/804); B65D (85/8043); B65D (85/8046); B65D (85/8049); B65D (85/8052); B65D (85/8055); B65D (85/8058); B65D (85/8061); B65D (85/8064); B65D (85/808); B65D (85/8085); B65D (85/812); B65D (85/816); B65D (85/84); B65D (88/005); B65D (88/02); B65D (88/12); B65D (88/1612); B65D (88/522); B65D (88/68); B65D (88/76); B65D (9/06); B65D (9/12); B65D (9/14); B65D (9/32); B65D (90/90006); B65D

(90/022); B65D (90/24); B65D (90/029); B65D (90/046); B65D (90/08); B65D (90/10); B65D (90/12); B65D (90/62); B65D (90/623)

USPC: 53/101; 53/115; 53/116; 53/121; 53/127; 53/129.1; 53/130.1; 53/131.1; 53/131.3; 53/133.2; 53/133.3; 53/133.4; 53/133.7; 53/136.1; 53/136.2; 53/136.3; 53/136.4; 53/136.5; 53/137.1; 53/137.2; 53/138.1; 53/138.2; 53/138.3; 53/138.6; 53/139.5; 53/140; 53/141; 53/145; 53/147; 53/148; 53/153; 53/155; 53/156; 53/157; 53/167; 53/168; 53/170; 53/171; 53/173; 53/176; 53/201; 53/202; 53/203; 53/209; 53/210; 53/211; 53/214; 53/216; 53/238; 53/239; 53/242; 53/243; 53/244; 53/245; 53/247; 53/248; 53/250; 53/252; 53/254; 53/255; 53/257; 53/262; 53/263; 53/267; 53/268; 53/272; 53/276; 53/278; 53/279; 53/280; 53/281; 53/282; 53/283; 53/284; 53/284.5; 53/285; 53/287; 53/288; 53/290; 53/292; 53/293; 53/294; 53/295; 53/296; 53/297; 53/298; 53/299; 53/300; 53/301; 53/302; 53/303; 53/305; 53/306; 53/307; 53/308; 53/309; 53/310; 53/311; 53/313; 53/314; 53/315; 53/317; 53/319; 53/320; 53/321; 53/324; 53/328; 53/329; 53/329.2; 53/329.3; 53/330; 53/331.5; 53/332; 53/333; 53/334; 53/338; 53/340; 53/341; 53/342; 53/343; 53/344; 53/345; 53/349; 53/350; 53/351; 53/352; 53/353; 53/354; 53/355; 53/356; 53/357; 53/359; 53/361; 53/363; 53/366; 53/367; 53/368; 53/369; 53/370.2; 53/370.3; 53/372.9; 53/374.4; 53/375.9; 53/376.4; 53/376.5; 53/376.7; 53/377.2; 53/377.6; 53/378.3; 53/381.1; 53/381.4; 53/382.1; 53/382.3; 53/383.1; 53/387.2; 53/389.1; 53/390; 53/391; 53/396; 53/397; 53/398; 53/399; 53/407; 53/409; 53/410; 53/411; 53/412; 53/413; 53/414; 53/415; 53/419; 53/420; 53/424; 53/425; 53/426; 53/427; 53/428; 53/430; 53/432; 53/434; 53/436; 53/438; 53/440; 53/441; 53/442; 53/443; 53/444; 53/445; 53/446; 53/447; 53/448; 53/449; 53/450; 53/451; 53/452; 53/453; 53/455; 53/456; 53/457; 53/458; 53/461; 53/462; 53/465; 53/467; 53/468; 53/469; 53/471; 53/472; 53/473; 53/474; 53/475; 53/477; 53/478; 53/48.1; 53/48.2; 53/48.4; 53/48.5; 53/48.7; 53/48.8; 53/48.9; 53/484; 53/485; 53/486; 53/487; 53/488; 53/489; 53/49; 53/490; 53/491; 53/492; 53/493; 53/495; 53/497; 53/502; 53/507; 53/510; 53/511; 53/52; 53/520; 53/521; 53/522; 53/525; 53/527; 53/528; 53/529; 53/531; 53/532; 53/537; 53/539; 53/541; 53/542; 53/543; 53/544; 53/55; 53/557; 53/558; 53/563; 53/564; 53/565; 53/566; 53/574; 53/578; 53/580; 53/582; 53/585; 53/587; 53/589; 53/590; 53/64; 53/67; 53/68; 53/70; 53/71; 53/72; 53/75; 53/76; 53/77; 53/88; 53/95; 53/DIG.2; 206/0.5; 206/1.5; 206/100; 206/102; 206/128; 206/139; 206/140; 206/141; 206/144; 206/147; 206/149; 206/15; 206/152; 206/153; 206/154; 206/155; 206/158; 206/161; 206/167; 206/19; 206/190; 206/194; 206/195; 206/196; 206/197; 206/198; 206/203; 206/204; 206/205; 206/210; 206/216; 206/217; 206/219; 206/221; 206/222; 206/223; 206/226; 206/229; 206/232; 206/233; 206/244; 206/246; 206/247; 206/254; 206/256; 206/259; 206/260; 206/264; 206/268; 206/270; 206/273; 206/276; 206/277; 206/291.3; 206/305; 206/306; 206/308.2; 206/310; 206/313; 206/315.1; 206/315.9; 206/318; 206/319; 206/320; 206/321; 206/322; 206/325; 206/326; 206/327; 206/343; 206/346; 206/349; 206/361; 206/363; 206/364; 206/365; 206/366; 206/37; 206/370; 206/372; 206/379; 206/38; 206/382; 206/384; 206/385; 206/386; 206/388; 206/389; 206/391; 206/392; 206/394; 206/395; 206/397; 206/398; 206/403; 206/407; 206/408; 206/410; 206/411; 206/413; 206/414; 206/415; 206/416; 206/418; 206/419; 206/422; 206/423; 206/424; 206/427; 206/429; 206/430; 206/431; 206/433; 206/434; 206/438; 206/442; 206/443; 206/445; 206/446; 206/447; 206/448; 206/449; 206/45.28; 206/451; 206/453; 206/454; 206/455; 206/457; 206/459.1; 206/459.5; 206/460; 206/461; 206/469; 206/470; 206/476; 206/477; 206/480; 206/482; 206/484; 206/484.2; 206/485; 206/486; 206/487; 206/488; 206/489; 206/490; 206/493; 206/497; 206/499; 206/5; 206/504; 206/506; 206/508; 206/509; 206/510; 206/514; 206/515; 206/517; 206/518; 206/519; 206/521; 206/522; 206/523; 206/524.1; 206/524.3; 206/524.4; 206/524.6;

206/524.7; 206/524.8; 206/525; 206/526; 206/527; 206/528; 206/529; 206/53; 206/530;
206/532; 206/534; 206/539; 206/541; 206/555; 206/557; 206/558; 206/562; 206/564;
206/568; 206/569; 206/571; 206/579; 206/581; 206/583; 206/585; 206/586; 206/587;
206/588; 206/589; 206/590; 206/591; 206/592; 206/593; 206/594; 206/596; 206/597;
206/598; 206/599; 206/6; 206/600; 206/68; 206/701; 206/702; 206/722; 206/723;
206/725; 206/758; 206/763; 206/769; 206/774; 206/775; 206/776; 206/777; 206/779;
206/780; 206/781; 206/782; 206/783; 206/784; 206/8; 206/800; 206/803; 206/804;
206/805; 206/806; 206/807; 206/812; 206/815; 206/818; 206/820; 206/822; 206/823;
206/83.5; 206/830; 206/831; 206/94; 206/96; 215/10; 215/11.1; 215/11.6; 215/12.1;
215/12.2; 215/14; 215/2; 215/200; 215/201; 215/203; 215/209; 215/216; 215/217;
215/220; 215/226; 215/227; 215/228; 215/229; 215/230; 215/231; 215/232; 215/233;
215/235; 215/237; 215/240; 215/244; 215/246; 215/248; 215/250; 215/251; 215/252;
215/253; 215/254; 215/256; 215/257; 215/258; 215/26; 215/260; 215/261; 215/270;
215/271; 215/272; 215/273; 215/274; 215/275; 215/276; 215/277; 215/278; 215/279;
215/285; 215/286; 215/287; 215/291; 215/292; 215/293; 215/295; 215/297; 215/298;
215/299; 215/30; 215/300; 215/302; 215/303; 215/304; 215/305; 215/306; 215/307;
215/308; 215/309; 215/310; 215/311; 215/313; 215/316; 215/317; 215/320; 215/321;
215/322; 215/323; 215/324; 215/325; 215/326; 215/328; 215/329; 215/330; 215/331;
215/332; 215/333; 215/334; 215/336; 215/337; 215/340; 215/341; 215/343; 215/344;
215/346; 215/347; 215/348; 215/349; 215/350; 215/351; 215/352; 215/353; 215/354;
215/355; 215/358; 215/359; 215/363; 215/364; 215/365; 215/371; 215/373; 215/376;
215/380; 215/382; 215/383; 215/385; 215/386; 215/387; 215/388; 215/390; 215/392;
215/395; 215/396; 215/397; 215/399; 215/40; 215/41; 215/42; 215/43; 215/44; 215/45;
215/46; 215/47; 215/48; 215/49; 215/50; 215/6; 215/901; 215/DIG.1; 215/DIG.2;
215/DIG.3; 215/DIG.7; 215/DIG.8; 220/200; 220/203.08; 220/203.29; 220/212;
220/212.5; 220/214; 220/229; 220/23.2; 220/23.4; 220/23.83; 220/23.87; 220/23.89;
220/23.9; 220/243; 220/254.1; 220/254.2; 220/254.3; 220/256.1; 220/257.1; 220/257.2;
220/258.2; 220/258.3; 220/258.5; 220/259.1; 220/259.3; 220/260; 220/265; 220/266;
220/268; 220/269; 220/270; 220/273; 220/274; 220/275; 220/277; 220/278; 220/279;
220/281; 220/282; 220/288; 220/289; 220/291; 220/293; 220/297; 220/303; 220/304;
220/310.1; 220/315; 220/316; 220/319; 220/323; 220/324; 220/326; 220/327; 220/328;
220/345.3; 220/351; 220/359.1; 220/359.2; 220/359.3; 220/359.4; 220/366.1; 220/367.1;
220/372; 220/375; 220/378; 220/380; 220/4.01; 220/4.15; 220/4.21; 220/4.24; 220/4.26;
220/4.28; 220/4.29; 220/495.01; 220/495.05; 220/495.06; 220/495.08; 220/495.11;
220/500; 220/506; 220/510; 220/519; 220/521; 220/522; 220/524; 220/560.01;
220/560.11; 220/560.15; 220/574; 220/574.1; 220/575; 220/578; 220/582; 220/586;
220/592.16; 220/592.17; 220/592.25; 220/592.26; 220/615; 220/618; 220/619; 220/62;
220/62.11; 220/62.12; 220/62.2; 220/62.22; 220/624; 220/627; 220/628; 220/631;
220/640; 220/641; 220/644; 220/651; 220/653; 220/658; 220/660; 220/662; 220/663;
220/669; 220/677; 220/703; 220/704; 220/705; 220/706; 220/707; 220/708; 220/709;
220/710; 220/711; 220/712; 220/713; 220/714; 220/716; 220/717; 220/718; 220/719;
220/721; 220/735; 220/737; 220/739; 220/740; 220/743; 220/757; 220/758; 220/763;
220/770; 220/775; 220/776; 220/780; 220/782; 220/783; 220/785; 220/786; 220/787;
220/788; 220/789; 220/790; 220/794; 220/796; 220/799; 220/8; 220/801; 220/803;
220/805; 220/810; 220/826; 220/835; 220/837; 220/839; 220/847; 220/86.1; 220/903;
220/906; 220/908; 220/910; 220/916; 220/DIG.15; 220/DIG.19; 220/1.5; 220/2.3 R;
220/201; 220/203.01; 220/203.09; 220/203.16; 220/210; 220/211; 220/213; 220/215;
220/221; 220/23.86; 220/23.91; 220/231; 220/234; 220/252; 220/253; 220/254.4;
220/254.5; 220/254.7; 220/254.8; 220/254.9; 220/255; 220/255.1; 220/258.1; 220/258.4;
220/259.2; 220/259.5; 220/262; 220/263; 220/264; 220/271; 220/276; 220/283; 220/284;

220/285; 220/287; 220/290; 220/295; 220/296; 220/298; 220/300; 220/301; 220/302;
220/305; 220/320; 220/321; 220/345.1; 220/345.2; 220/345.4; 220/345.6; 220/348;
220/350; 220/360; 220/363; 220/368; 220/371; 220/373; 220/374; 220/4.02; 220/4.12;
220/4.22; 220/4.23; 220/4.27; 220/4.33; 220/475; 220/481; 220/482; 220/484; 220/495;
220/495.02; 220/495.03; 220/495.04; 220/495.07; 220/495.1; 220/503; 220/507; 220/508;
220/515; 220/523; 220/526; 220/528; 220/529; 220/530; 220/531; 220/532; 220/551;
220/555; 220/556; 220/560.02; 220/560.13; 220/565; 220/566; 220/567.1; 220/567.2;
220/568; 220/573.1; 220/573.2; 220/574.3; 220/590; 220/592.15; 220/592.2; 220/592.24;
220/592.27; 220/6; 220/600; 220/601; 220/602; 220/609; 220/612; 220/613; 220/617;
220/62.15; 220/62.18; 220/62.19; 220/620; 220/621; 220/623; 220/625; 220/626;
220/643; 220/645; 220/646; 220/652; 220/657; 220/659; 220/661; 220/666; 220/670;
220/671; 220/674; 220/675; 220/676; 220/678; 220/692; 220/694; 220/695; 220/697.7;
220/700; 220/710.5; 220/715; 220/720; 220/724; 220/727; 220/729; 220/730; 220/733;
220/738; 220/746; 220/751; 220/752; 220/756; 220/771; 220/772; 220/784; 220/791;
220/792; 220/793; 220/802; 220/806; 220/811; 220/812; 220/815; 220/821; 220/825;
220/827; 220/829; 220/830; 220/831; 220/832; 220/833; 220/834; 220/836; 220/838;
220/840; 220/841; 220/844; 220/845; 220/848; 220/849; 220/86.2; 220/87.1; 220/89.1;
220/902; 220/908.1; 220/908.2; 220/908.3; 220/909; 220/913; 220/915; 220/915.1;
220/919; 220/DIG.13; 220/DIG.2; 220/DIG.20; 220/DIG.22; 220/DIG.27; 220/DIG.32;
220/DIG.33; 220/DIG.9; 220/FOR.173; 229/125.03; 229/87.06; 493/100; 493/101;
493/102; 493/103; 493/104; 493/105; 493/106; 493/107; 493/108; 493/109; 493/110;
493/111; 493/114; 493/115; 493/116; 493/119; 493/12; 493/120; 493/121; 493/122;
493/126; 493/128; 493/129; 493/131; 493/132; 493/133; 493/136; 493/14; 493/141;
493/142; 493/144; 493/148; 493/149; 493/150; 493/152; 493/153; 493/154; 493/155;
493/156; 493/158; 493/159; 493/162; 493/163; 493/164; 493/165; 493/167; 493/171;
493/175; 493/176; 493/177; 493/180; 493/181; 493/183; 493/184; 493/185; 493/210;
493/211; 493/214; 493/218; 493/219; 493/225; 493/231; 493/232; 493/253; 493/264;
493/270; 493/274; 493/276; 493/279; 493/285; 493/287; 493/288; 493/292; 493/295;
493/296; 493/297; 493/298; 493/303; 493/309; 493/313; 493/315; 493/316; 493/317;
493/318; 493/319; 493/324; 493/329; 493/33; 493/331; 493/332; 493/336; 493/337;
493/338; 493/340; 493/343; 493/344; 493/372; 493/377; 493/379; 493/38; 493/380;
493/381; 493/385; 493/388; 493/39; 493/390; 493/397; 493/407; 493/416; 493/418;
493/429; 493/441; 493/442; 493/449; 493/453; 493/454; 493/460; 493/464; 493/465;
493/466; 493/468; 493/472; 493/474; 493/475; 493/52; 493/55; 493/56; 493/59; 493/61;
493/63; 493/66; 493/73; 493/74; 493/75; 493/76; 493/77; 493/79; 493/82; 493/83;
493/84; 493/85; 493/87; 493/88; 493/89; 493/907; 493/929; 493/93; 493/938; 493/941;
493/95; 493/956; 493/959; 493/962; 493/964; 493/966; 493/967; 493/968; 493/98; 493/99

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2120903	12/1937	Arlington	493/158	B65D 41/04
2233489	12/1940	Reifsnyder	N/A	N/A
2505748	12/1949	Barbieri	229/5.5	B65D 3/10
2746669	12/1955	Arlington	229/125.17	B29C 66/12441

6196451	12/2000	Helms	229/5.5	B29C 45/14336
2004/0144786	12/2003	Chmela	220/257.1	B65D 55/0863
2008/0099481	12/2007	D' Amato	220/276	B65D 43/0262
2011/0233208	12/2010	Kim	220/254.1	B65D 43/0212

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103482207	12/2017	CN	B65D 45/32
110979960	12/2019	CN	B65D 45/02
321716	12/1928	GB	N/A
2004001892	12/2003	JP	B65D 43/02
20220020505	12/2021	KR	B65D 43/0212

OTHER PUBLICATIONS

Search Report issued in corresponding European Application No. 23156018.6, with English translation of categories of cited documents, dated Aug. 10, 2023 (9 pages). cited by applicant

Primary Examiner: Stashick; Anthony D

Assistant Examiner: Molina; Marcos Javier Rodriguez

Attorney, Agent or Firm: FLYNN THIEL, P.C.

Background/Summary

TECHNICAL FIELD

(1) The invention relates to a lid from paper, paperboard or paper-like material for a container, wherein the lid is produced from an integral planar blank and has a lid plate and a lid collar, wherein the lid collar proceeds from a periphery of the lid plate, wherein the lid collar is formed from at least two plies of the planar blank, wherein a first ply of the lid collar, proceeding from the periphery of the lid plate, extends to a periphery of the lid collar, and wherein a second ply of the lid collar, proceeding from the periphery of the lid collar, extends along a radially inner side of the lid collar and terminates at a second ply peripheral end portion.

BACKGROUND AND SUMMARY

(2) A lid can be provided for latching to a container having a so-called mouth roll. In lids from paper, paperboard or paper-like material for a container, which are produced from an integral planar blank, it is problematic that the second, inner ply of the lid collar has to be sealed to the outer ply of the lid collar by means of an adhesive or a plastics-material coating, such that the lid collar maintains its shape. Plastics-material coatings here are problematic from an ecological point of view.

(3) A lid from paper, paperboard or paper-like material for a container, and a method for producing a lid, are to be improved by the invention such that producing the lid from paper, paperboard or paper-like material which is free of plastics material is possible.

(4) In a lid from paper, paperboard or paper-like material for a container it is provided that the lid is

produced from an integral, planar blank and has a lid plate and a lid collar, that the lid collar proceeds from a periphery of the lid plate, that the lid collar is formed from at least two plies of the planar blank, that a first ply of the lid collar, proceeding from the periphery of the lid plate, extends to a periphery of the lid collar, and that a second ply of the lid collar extends from the periphery of the lid collar on a radially inner side thereof to a second ply peripheral end portion, wherein the second ply peripheral end portion at least in portions is received in a groove in a region of the periphery of the lid plate.

(5) In that the second ply peripheral end portion of the lid collar at least in portions is received in a groove in the region of the periphery of the lid plate, the peripheral end portion is mechanically fixed in the groove; in other words, the flap of the lid collar that is formed by the second ply latches into the groove. The second ply of the lid collar is fixed solely by the latching action. As a result, it is possible for the lid according to the invention not to be produced from coated paper, paperboard or paper-like material, or it is possible to completely dispense with adhesive in the production of the lid.

(6) The lid according to the invention is produced from paper, paperboard or paper-like material. Such lids are usually also referred to as paperboard lids. A material which can be processed like paper or paperboard is referred to as a paper-like material. Paper or paperboard comprises fibers and consequently cannot be elongated in practical terms and also be compressed only to a very minor extent. In particular, paper, paperboard or paper-like material cannot be vacuum-formed or heavily deformed in general. Plastics material which cannot be stretched, or stretched only to a very minor extent, is also considered to be paper-like material, for example. In the production of a lid from such a paper-like plastics material, the lid has to be produced from a planar blank which is then deformed to the extent permitted by the paper-like plastics material. In other words, the lid according to the invention from paper, paperboard or paper-like material is always produced in the same way, specifically by deforming an integral planar blank.

(7) The lid according to the invention can be configured as a so-called snap-on lid or else as a so-called snap-in lid. In a snap-on lid, the internal circumference of the lid collar is larger than the external circumference of the upper periphery of a container onto which the lid is then placed. In the state placed thereon, the lid collar then surrounds the external wall of the container on the upper end of the latter. In a snap-in lid, the external circumference of the lid collar is slightly smaller than the internal circumference of the container on the upper end of the latter. The lid collar is thus snapped into the upper end of the container. In the state placed thereon, the external wall of the container by way of the upper region of the former then surrounds the lid collar at least in portions.

(8) In a refinement of the invention, the groove in the lid plate is configured to be encircling.

(9) In this way, the periphery of the second, inner ply of the lid collar can be latched and fixed across the entire circumference of the lid plate.

(10) In a refinement of the invention, the lid plate has an upper side and a lower side, wherein the second ply of the lid collar from the periphery of the lid collar extends toward the lower side of the lid plate, and an encircling protrusion is disposed on the upper side of the lid plate, wherein the groove is disposed on a side of the protrusion that faces the lower side of the lid plate.

(11) The groove for receiving the peripheral end portion of the second ply of the lid collar is created simultaneously on the lower side of the lid plate while shaping the encircling protrusion on the upper side of the lid plate. For example, the encircling protrusion, and thus the encircling groove, can be embossed or curled in the planar blank.

(12) In a refinement of the invention, the first ply of the lid collar forms a wall of the groove.

(13) In this way, the groove is disposed directly on the outer periphery and thus at the transition from the lid plate to the lid collar. In the fixed state, the first ply and the second ply of the lid collar can therefore lie directly on top of each other such that the lid collar can be configured to be very stable and with double the wall thickness of the planar blank.

(14) In a refinement of the invention, the groove is spaced apart from the periphery of the lid plate.

(15) In the context of the invention it may be expedient for the groove to be spaced apart from the periphery of the lid plate. In this case, the lid collar is still composed of two plies of the planar blank, said plies however being mutually spaced apart on the lower side of the lid plate.

(16) In a refinement of the invention, the lid plate has an upper side and a lower side, wherein the second ply of the lid collar, from the periphery of the lid collar, extends in the direction of the lower side of the lid plate, wherein a protrusion which extends away from the lid plate is disposed on the lower side of the lid plate, wherein the protrusion is spaced apart from the periphery of the lid plate, and wherein the groove is formed between the protrusion and the first ply of the lid collar.

(17) For example, the protrusion can be formed in that embossing is performed from the upper side of the lid plate, said embossing then configuring a protrusion on the lower side of the lid plate. The peripheral end portion of the second ply of the lid collar can then be moved across this protrusion and latch behind this protrusion. The groove is then formed between the protrusion and the first ply of the lid collar.

(18) In a refinement of the invention, the peripheral end portion of the second ply of the lid collar is secured in the groove by means of an adhesive.

(19) Fixing the peripheral end portion of the second ply of the lid collar in the groove can be improved in this way.

(20) The adhesive is configured as a hot-melt adhesive, for example.

(21) In a refinement of the invention, the planar blank from which the lid is produced is not provided with a sealable plastics-material coating.

(22) As a result of the mechanical fixing of the peripheral end portion of the second ply of the lid collar in the groove, a sealable plastics-material coating can be dispensed with. As a result, the lid according to the invention can be produced in a very ecologically friendly way and can readily be produced from paper material which is rapidly biologically degradable, for example.

(23) In a refinement of the invention, the lid collar on the radially inner side thereof has at least one second protrusion which extends radially inward.

(24) By means of such a protrusion, the lid collar can latch onto a cup with a mouth roll, or an encircling protrusion which protrudes radially outward. In this instance, the mouth roll lies between the second protrusion and the lower side of the lid plate of the lid. As a result, the lid is securely held on the container.

(25) In a refinement of the invention, the second protrusion is configured as a corrugation in the first ply and/or of the second ply of the lid collar.

(26) Such a corrugation can be curled or embossed in the lid collar, for example, and is in particular adapted to the dimensions of a mouth roll. For example, the corrugation can also be configured such that the corrugation receives a portion of the mouth roll of the container.

(27) In a refinement of the invention, a convex side of the corrugation lies radially on the inside.

(28) In this way, a second protrusion which in the radial direction protrudes inward is formed by means of the corrugation, wherein the mouth roll in this instance can be received between the second protrusion and the lower side of the lid plate.

(29) In a refinement of the invention, the second protrusion is a thickening of the second ply of the lid collar.

(30) A thickening can be achieved by compressing the second ply, for example. A thickening can however also be achieved by curling a corrugation and subsequently compressing the corrugation, for example.

(31) In a refinement of the invention, the periphery of the second ply of the lid collar is provided with clearances or is configured so as to be wave-shaped.

(32) Such a design of the periphery of the second ply minimizes material stresses when folding over the second ply. In this instance, the periphery of the second ply of the lid collar is inevitably fixed in the groove only in portions.

(33) In a refinement of the invention, the second ply of the lid collar has incisions which proceed

from the periphery of the second ply, wherein portions disposed between the incisions are disposed so as to at least in portions run obliquely radially inward.

(34) In this way, barb-type elements on the lid collar can be formed by way of the portions that run obliquely inward. The lid can then be pushed onto a mouth roll until the obliquely disposed portions snap in behind the mouth roll. The lid is then securely fixed on the container. The portions can be configured such that the lid is either placed so as not to be releasable in a non-destructive manner, or is placed as to be releasable upon overcoming specific resistance.

(35) In a refinement of the invention, the lid collar, in particular on the periphery thereof that is spaced apart from the lid plate, has at least one protrusion that extends radially outward.

(36) In a snap-in lid, such a protrusion that extends radially outward is expedient for forming a detent which defines a maximum depression depth of the lid when inserting the latter into a container. The protrusion can encircle the entire circumference of the lid, but this is not mandatory in the context of the invention. The encircling protrusion that extends radially outward can be established in a simple manner, for example, by forming the lid collar by bending the latter radially outward by 90°. For example, two mutually opposite protrusions which extend in each case across an angle of approximately 100° to 150°, in particular 130° to 140°, about the circumference of the lid collar are provided. An extraction aid for the lid can then be disposed between these two radially outward extending protrusions, for example.

(37) In a refinement of the invention, the lid collar, in particular on the periphery thereof that is spaced apart from the lid plate, has at least one protrusion that extends radially inward.

(38) In this way, an extraction aid for a user can be provided in a very simple manner, in particular in a snap-in lid. The user reaches into the lid and places one or a plurality of fingers below the radially inward protruding protrusion. As a result, the user can extract the lid in a simple manner by way of an upward movement.

(39) The radially inward protruding protrusion can extend across the entire circumference of the lid. For example, a first protrusion that extends radially outward and defines a depression depth of the lid can be provided, and a second protrusion that extends radially inward and in the height direction is disposed at a different position that the radially outward extending protrusion can be provided. However, the radially inward extending protrusion advantageously extends only across a sub-region of the circumference, and at least one radially outward extending protrusion is provided at the same height in a different sub-region of the circumference. For example, two mutually opposite and radially inward extending protrusions are provided. The radially inward extending protrusions here can extend across the circumference of the lid by an angle between 30° and 60°, in particular 45° to 55°.

(40) The object on which the invention is based is also achieved by a method for producing a lid having a lid plate and a lid collar which proceeds from the periphery of the lid plate. The method includes configuring a pot-type intermediate element from a planar blank from paper, paperboard or paper-like material; folding over an encircling wall assembly of the pot-type element such that a first ply of the lid collar forms in the radial direction, an outside of the lid collar, and a second ply of the lid collar forms an inside of the lid collar in the radial direction; forming a groove in a radially outer region of the base of the pot-type intermediate element; and latching at least one portion of the second ply peripheral end portion into the groove.

(41) Using the method according to the invention, an integral lid from paper, paperboard or paper-like material which has a two-ply stable lid collar and can nevertheless be produced without the use of an adhesive or a sealable coating can be produced. This because the peripheral end portion of the inner second ply of the lid collar latches in the groove and is mechanically fixed as a result. The method according to the invention can also readily be used together with an adhesive, for example a hot-melt adhesive, or coated paper material. There are paper materials which are provided with a plastics-material coating and provide only very minor cohesion when sealing. An integral lid can also be produced by way of the method according to the invention using such materials. In the

context of the invention, the planar material for producing the lid can also be provided with a sealable dispersion coating and/or with a protective varnish against invading liquids. For example, the fixing of the peripheral end portion of the second ply of the lid collar in the groove can also be improved in that the peripheral end portion is fixed in the groove by way of an adhesive. This can be advantageous when a very stable lid is to be produced, for example. However, even when the periphery of the second ply of the lid collar is fixed by way of an adhesive, the utilization of an adhesive thus only assists the mechanical fixing of the peripheral end portion of the second ply of the lid collar in the groove, as is already present according to the invention.

(42) In the refinement of the invention the shaping of the groove is provided in a way that a portion of the first ply of the lid collar forms a lateral wall of the groove.

(43) In this way, the groove is disposed directly on the periphery of the lid collar, and the second ply of the lid collar can bear in a planar manner on the first ply of the lid collar.

(44) In the refinement of the invention the shaping of the groove is provided in a way that an encircling protrusion is configured on an upper side of the lid plate that lies opposite the lid collar, wherein the groove is disposed on a side of the protrusion facing the lower side of the lid plate.

(45) In the refinement of the invention the shaping of the groove is provided in a way that the protrusion forms the outer periphery of the lid plate.

(46) In this way, the lid plate is stabilized and an appealing design of the lid is also derived.

(47) In the refinement of the invention, fixing at least portions of the peripheral end portion of the second ply in the groove by means of an adhesive is provided.

(48) In this way, materially integral fixing in addition to the mechanical fixing of the peripheral end portion of the second ply in the groove can be provided.

(49) In a refinement of the invention, the shaping of the lid plate by means of embossing the base of the pot-type intermediate element is provided.

(50) In this way, the lid plate can be stabilized by embossings or stampings. For example, it is also possible for the lid plate to be provided with an insertion opening or a drinking opening, and for embossings or stampings to be provided in order to enable drinking from the container while the lid is placed on the latter. For example, an embossing can be provided such that liquid which has potentially made its way to the upper side of the lid plate runs back into the container.

(51) Further features and advantages of the invention are derived from the claims and the description hereunder of preferred embodiments of the invention in conjunction with the drawings. Individual features of the various embodiments illustrated and described can be combined with one another in an arbitrary manner without departing from the scope of the invention. This also applies to the combination of individual features without other individual features, in connection with which they are illustrated and/or described. In the drawings:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a lid according to the invention and according to a first embodiment from obliquely below;

(2) FIG. 2 shows a sectional view of the lid of FIG. 1 from obliquely above;

(3) FIG. 3 shows the detail III from FIG. 2 in an enlarged illustration;

(4) FIG. 4 shows a lid according to the invention and according to a second embodiment from obliquely below;

(5) FIG. 5 shows a sectional view of the lid of FIG. 4 from obliquely above;

(6) FIG. 6 shows the detail VI from FIG. 5 in an enlarged illustration;

(7) FIG. 7 shows an inventive lid according to a third embodiment of the invention from obliquely below;

- (8) FIG. 8 shows a sectional view of the lid of FIG. 7 from obliquely above;
- (9) FIG. 9 shows the detail IX from FIG. 8 in an enlarged illustration;
- (10) FIG. 10 shows a lid according to the invention and according to a fourth embodiment from obliquely below;
- (11) FIG. 11 shows a sectional view of the lid of FIG. 10 from obliquely above;
- (12) FIG. 12 shows the detail XII from FIG. 11 in an enlarged illustration;
- (13) FIG. 13 shows a sectional view of the lid of FIG. 10 in the state placed on a paperboard cup with a mouth roll from obliquely above;
- (14) FIG. 14 shows the detail XIV from FIG. 13 in an enlarged illustration;
- (15) FIG. 15 shows the cup and the lid of FIG. 13 in a sectional view from slightly obliquely below;
- (16) FIG. 16 shows the detail XVI from FIG. 15 in an enlarged illustration;
- (17) FIG. 17 shows a sectional view of a cup according to the invention and according to fifth embodiment from obliquely above;
- (18) FIG. 18 shows the detail XVIII from FIG. 17 in an enlarged illustration;
- (19) FIG. 19 shows a sectional view of a lid according to the invention and according to a sixth embodiment from obliquely below;
- (20) FIG. 20 shows the detail XX from FIG. 19 in an enlarged illustration;
- (21) FIG. 21 shows a view of a lid according to the invention and according to a seventh embodiment from obliquely above;
- (22) FIG. 22 shows a plurality of the lids from FIG. 21 in the stacked state;
- (23) FIG. 23 shows a section view of the stacked lids from FIG. 22;
- (24) FIG. 24 shows a view of a lid according to the invention and according to an eighth embodiment from obliquely above;
- (25) FIG. 25 shows a sectional view of the lid of FIG. 24;
- (26) FIG. 26 shows the enlarged detail A from FIG. 25;
- (27) FIG. 27 shows a view of a lid according to the invention and according to a ninth embodiment from obliquely above;
- (28) FIG. 28 shows the lid of FIG. 27 in the state placed on a container;
- (29) FIG. 29 shows a plan view onto the lid of FIG. 27;
- (30) FIG. 30 shows a view onto the section plane A-A in FIG. 29;
- (31) FIG. 31 shows the detail A from FIG. 30;
- (32) FIG. 32 shows portions of a sectional view of the container with the lid from FIG. 28;
- (33) FIG. 33 shows portions of a further sectional view of the container with the lid from FIG. 28;
- (34) FIG. 34 shows a view of a lid according to the invention and according to a tenth embodiment from obliquely above;
- (35) FIG. 35 shows the lid of FIG. 34 in the state placed on a container;
- (36) FIG. 36 in portions shows a sectional view of the lid with the container from FIG. 35; and
- (37) FIG. 37 in portions shows a further sectional view of the lid with the container from FIG. 35.

DETAILED DESCRIPTION

(38) The illustration of FIG. 1 shows a lid 10 according to the invention and according to a first embodiment of the invention. The lid 10 has a lid plate 12 and lid collar 14 which protrudes from the lid plate 12. In the lid 10, the lid collar 14 protrudes from the lid plate 12 downward at an angle of 90°. Only the lower side of the lid plate 12 is visible in FIG. 1. The lid 10 is provided for placing on a container, usually a paperboard cup with a mouth roll, cf. FIG. 13 and FIG. 15. In the placed state, an upper side of the mouth roll of the paperboard cup bears on the lower side of the lid plate 12, and an inside of the lid collar 14 bears on an outside of the mouth roll.

(39) As can be seen from FIG. 1 and, for example, also from FIGS. 13 to 15, the lid 10 is configured as a so-called snap-on lid which is pushed over the upper end of a container, or of a sleeve, and especially over the mouth roll of a container. A lid which is configured as a so-called

snap-in lid may also be provided in the context of the invention, see FIGS. 21 to 37. This type of lid is not pushed over the sleeve or the mouth roll of a container, but is inserted into a container. In order for the lid configured as a snap-in lid not to slide into the container, a holding element is typically situated on the outside of said lid. In the design of a snap-in lid from paper, the question is where the cutting edge of the planar blank, the so-called round tops, is to be positioned so as not to have any contact with the filling material, on the one hand, and not to expose the cutting edge so as not to be protected in relation to external environmental influences, on the other hand. Also in the case of a snap-in lid the invention offers the advantage that the periphery or peripheral end portion of the second ply of the lid collar, at least in portions, is received in a groove in the region of the periphery of the lid plate such that the periphery of the second ply is mechanically locked, on the one hand, and simultaneously covered and protected, on the other hand. In a snap-in lid without any plastics-material coating, the support of the preformed border, or of the external circumference of the lid collar, on the internal wall of the container is advantageous because the border, or the lid collar, is less dimensionally stable as a result of the absence of the plastics-material coating.

(40) In the context of the invention, the lid **10** can be embodied so as to be circular, as illustrated in FIG. 1. Lids having a non-circular cross section, for example rectangular with rounded ends, oval, elliptic or the like are also within the scope of the invention.

(41) The lid plate **12** has a central embossing **16** which is slightly depressed in comparison to the peripheral region **18** of the lid plate **12**, for example depressed by 1 mm to 2 mm in the embodiment illustrated. The embossing **16** serves for reinforcing the lid plate **12**. Embossings with a different shape, for example embossings for implementing ribs, webs or the like, may also be provided instead of the depression **16**.

(42) The lid plate **12** can also be provided with a drinking opening. For example, embossings or stampings can also be provided in the lid plate **12**. For example, a drinking opening can be disposed at the lowest point of an embossing. Stampings and/or embossings can also be provided in the region of a drinking opening in order to render drinking from the drinking opening comfortable and to ensure that potential liquid on the upper side of the lid plate **12** runs back into the container.

(43) The lid **10** is produced from an integral planar blank, as will yet be explained hereunder. The integral blank is composed of paper, paperboard or paper-like material. Such lids are usually also referred to as paperboard lids. Material that can be processed like paper or paperboard is referred to as paper-like material. Paper or paperboard comprises fibers and can consequently not be elongated in practical terms, and also be compressed only to a very minor extent. Paper, paperboard or paper-like material can in particular not be vacuum-formed or heavily deformed in general. The embossing **16**, and optionally embossings or stampings in the region of a drinking opening or for reinforcing the lid **10** can, however, be incorporated when said embossings or stampings have only very minor dimensions, like the embossing **16** with a depth of only 1 mm to 2 mm in the embodiment illustrated. Plastics material which cannot be stretched, or stretched only to a very minor extent, is also considered to be paper-like material, for example. In the production of a lid from such a paper-like plastics material, the lid has to be produced from a planar blank.

(44) Conventional lids from paper or paperboard are either configured in two parts, wherein the lid collar and the lid plate in this instance are adhesively bonded to each other or sealed to each other by means of a skirt. Integral lids in the region of the lid collar are typically embodied with two plies in order to be sufficiently stable. Connecting an inner ply to an outer ply likewise takes place by sealing a plastics-material layer on the surfaces of the planar blank. Such plastics-material layers indeed permit two plies to be sealed and thus two plies of the planar blank to be fixed to one another. However, such plastics-material layers are problematic in recycling because the lid in this instance is composed of two dissimilar materials. The biological degradability of the material from which the lid is produced is also compromised or even completely prevented. In contrast, the lid **10** according to the invention can be produced from non-coated paper, non-coated paperboard, or non-coated paper-like material and without the utilization of an adhesive, as is still to be explained

hereunder. However, in the context of the invention the use of plastics-material coated paper, plastics-material coated paperboard, or paper-like material coated with adhesive or sealing material is not precluded. In the context of the invention, a planar blank from paper, paperboard or paper-like material, which for sealing is provided with a dispersion coating and/or a protective varnish against invading liquids, can also be used.

(45) FIG. 2 shows a sectional view of the lid **10** of FIG. 1 from obliquely above. It can be seen in this view that the lid **10** is produced from an integral, planar blank and that the lid plate **12** is composed of one ply of the planar blank, and the lid collar **14** is composed of two plies of the planar blank. An encircling protrusion **20** having an approximately semicircular cross section can be seen on the outermost periphery of the lid plate **12**. The planar blank is deformed in order for this protrusion to be formed. This can be performed by embossing from the lower side of the lid plate **12**, cf. also FIG. 3, or optionally by suitably compressing the planar blank. Adjoining the protrusion **20**, a first ply **22** of the lid collar **14** extends away from the lid plate **12**. In the embodiment illustrated, the first ply **22** extends downward, so as to be perpendicular to the lid plate **12**. The planar blank is folded over by 180° on the periphery **24** of the lid collar **14**, which in the embodiment of FIGS. 1 to 3 forms the lower periphery of the lid collar **14** and thus has a maximum spacing from the lid plate **12**. Proceeding from the periphery **24** of the lid collar **14**, a second ply **26** of the lid collar **14** extends in the direction toward the lid plate **12** again. In the embodiment illustrated, the second, inner ply **26** bears in a planar manner on the first, outer ply **22** of the lid collar **14**. The second ply **26** extends up to or terminates at a peripheral end portion. This peripheral end portion of the second ply **26** is received in a groove **28** which is formed on the lower side of the protrusion **20**, see also FIG. 3. The groove is thus accessible from the lower side of the lid plate **12** and encircles the entire periphery of the lid plate **12**. The second ply **22** in terms of length is configured such that the peripheral end portion of the second ply **26**, and thus the upper delimitation of the second ply **26** in FIGS. 2 and 3, can be pressed into the groove **28** only counter to a resistance. Consequently the second ply **26** is folded over by 180° on the lower periphery **24** and latches in the groove **28** by pressure. The position of the second ply **22** is fixed as soon as the peripheral end portion of the second ply **26** is disposed in the groove **28**, as is illustrated in FIGS. 2 and 3. No adhesive is required in order for this fixing to be achieved. Rather, the peripheral end portion of the second ply **26** is held in the position illustrated in FIGS. 2 and 3 as a result of the disposal of the former in the groove **28**. As a result the lid **10** according to the invention can be produced without the utilization of an adhesive.

(46) In the context of the invention, the fixing of the second ply **26**, and especially of the peripheral end portion of the second ply **26**, in the groove **28** can also be assisted by adhesive bonding in the groove **28**, or else by planar or partial sealing of the second ply **26** to the first ply **22** of the lid collar **14**.

(47) The illustration of FIG. 4 shows a lid **40** according to the invention and according to a second embodiment of the invention from obliquely below. Functionally equivalent elements, or even elements of identical construction, of the lid **40** are provided with the same reference signs as in the lid of FIGS. 1 to 3, and are not explained once again. The lid **40** is produced from an integral planar blank and has a lid plate **12** and a lid collar **14**. Proceeding from an encircling protrusion **20** on the periphery of the lid plate **12**, the lid collar **14** extends downward. A first ply **22** of the lid collar **14** adjoins the protrusion **20**. The planar blank is folded over by 180° on the periphery **24** of the lid collar, and a second ply **26** of the lid collar extends in the direction toward the lower side of the lid plate **12** again. The peripheral end portion of the second ply **22** is again received in the groove **28** on the lower side of the protrusion and fixed as a result, see FIG. 6.

(48) It can be seen in FIG. 3 and FIG. 6 that the first ply **22** forms the outer lateral delimitation of the groove **28**.

(49) The inner ply **26** is provided with a thickened portion **42**. This thickened portion **42** encircles the entire lid collar **14** and forms a protrusion that protrudes radially inward. The thickened portion

42 is produced, for example, in that a groove is molded in the second ply **26** and this groove is then grouted in order for the thickened portion **42** to be produced. The thickened portion **42** can also be produced by compressing the second ply **26**. The thickened portion **42** has a predefined spacing from the lower side of the lid plate **12**. This predefined spacing is adapted to the diameter of a mouth roll of a container. As a result, the mouth roll of a container can be received between the lower side of the lid plate **12** and the upper side of the thickened portion **42** that faces the lid plate **12**. By means of the thickened portion **42**, the lid **30** can latch onto the mouth roll of a container, for example of a paperboard container.

(50) FIG. 7 shows a lid **50** according to the invention and according to a third embodiment of the invention in a view from obliquely below. Elements of the lid **50** which are functionally equivalent, or even elements of identical configuration, to the lid **10** of FIGS. 1 to 3 are provided with the same reference signs, and are not explained once again. The lid **50** has a lid plate **12** and a lid collar **14** which protrudes from the periphery of the lid plate **12** so as to be perpendicular to the latter. As can be seen in FIGS. 8 and 9, an encircling protrusion **20** is configured on the periphery of the lid plate **12**, a groove **28** in which the periphery of a second, inner ply **26** of the lid collar **26** is able to be received being disposed on the lower side of said protrusion **20**. The first, outer ply **22** is folded over by 180° on the periphery **24** of the lid collar **14** such that the second, inner ply **26** of the lid collar extends in the direction toward the lid plate **12** again.

(51) The lid collar **14** is provided with an encircling corrugation **52** which forms a protrusion **54** that protrudes radially inward. The corrugation **52** is embossed, for example curled, into the lid collar **14** from a radially outer side. The corrugation **52** is thus configured so as to be concave on the radially outer side of the lid collar **14**. The corrugation **52** is configured so as to be convex on the radially inner side of the lid collar **14**, and as a result forms the protrusion **54** that protrudes radially inward. The corrugation is incorporated in both plies **24**, **26** of the lid collar **14**. In the context of the invention, a corrugation may also be incorporated only in the second, inner ply **26** of the lid collar **14**.

(52) The inward-protruding protrusion **54** is disposed at a spacing from the lower side of the lid plate **12**. As a result, the mouth roll of a container, for example of a paperboard container, can be received between the lower side of the lid plate **12** and the upper side of the protrusion **54** that faces the lower side of the lid plate **12**. As a result, the lid **50** can latch onto a container with a mouth roll.

(53) Owing to the inward-protruding protrusion **54** being configured in the shape of a segment of a circle, a lid can in this instance be easily removed again, applying only a minor force, from a container with a mouth roll.

(54) FIG. 10 shows a lid **60** according to the invention and according to a fourth embodiment of the invention. Elements of the lid **60** which are functionally equivalent, or even elements which are identical, to the lid **10** of FIGS. 1 to 3 are provided with the same reference signs, and are not explained once again.

(55) The lid **60** has a lid plate **12** and a lid collar **14** which protrudes from the lid plate **12**. An encircling protrusion **20**, a groove **28** being disposed on the lower side thereof, is disposed on the upper side of the lid plate **12**, see FIGS. 11 and 12. A first ply **22** of the lid collar **14**, which is disposed radially on the outside, forms the radially outer lateral wall of the groove **28**. The inner ply **26** of the lid collar **14** is folded over by 180° on the periphery **24** of the lid collar **14** and again extends in the direction toward the lower side of the lid plate **12**. A periphery of the second, inner ply **26** of the lid collar **14** is in portions received in the groove **28** and as a result fixed in the position thereof illustrated in FIGS. 10 to 12.

(56) The second ply **26** is provided with a plurality of incisions **62** which are disposed so as to be mutually parallel and, so as to be perpendicular to the periphery of the second ply **26**, run in the direction toward the periphery **24** of the lid collar **14**. Material portions that lie between two incisions **62** are either disposed so as to be parallel to the first ply **22** and, by way of the periphery of said material portions, received in the groove **28**, or portions **64** between two adjacent incisions

62 are shortened and protrude radially obliquely inward. As can be readily seen in FIG. **12**, the portions **64** protrude obliquely inward in the direction toward the lower side of the lid plate **12** and, as a result, form barb-type elements. Portions **64** are distributed about the entire circumference of the lid collar **14**, on the radially inner side thereof, and are uniformly spaced apart from one another.

(57) When the lid **60** is pushed onto a mouth roll of a container, for example of a paperboard container, the portions **64** are initially deflected outward until the latter are disposed so as to be approximately parallel to the first, outer ply **22** of the lid collar **14**. As soon as the mouth roll then impacts the lower side of the lid plate **12**, the portions **64** can spring back to the position shown in FIGS. **10** to **12**. As a result, the lid **60** is then securely fastened to the mouth roll of the container. This state can be readily seen in FIGS. **13** to **16**. It can also be readily seen in FIGS. **13** to **16** that the lid **60** typically cannot be removed from the container **70** again in a non-destructive manner. At least some of the portions **64** will be completely bent back or possibly even torn off when the container is removed. As a result, it can be ensured that the lid **60** is very securely held on the mouth roll **72** of the cup **70**.

(58) The illustration of FIG. **17** shows a lid **80** according to the invention and according to a fifth embodiment of the invention, said lid **80** being of a construction substantially identical to that of the lid **40** of FIGS. **4** to **6**. Therefore, only the points of differentiation in comparison to the lid **40** of FIGS. **4** to **6** will be explained.

(59) As opposed to the lid **40** of FIGS. **4** to **6**, the periphery of the second ply **26** of the lid collar **14** in the lid **80** is secured on the lower side of the protrusion **20** by means of an adhesive bead **82** in the groove **28**, see FIG. **18**. The adhesive bead **82** may be composed of a hot-melt adhesive, for example, and be incorporated in the groove **28** before the second ply **26** is completely folded over by 180° on the periphery **24** of the lid collar **14**. As soon as the periphery of the second ply **26** has been pushed into the groove **28**, the adhesive bead **82** solidifies and, as a result, reliably secures the periphery of the second ply **26** in the groove **28**. As a result, the lid **80** can be configured so as to be even more stable in comparison to the lid **40**.

(60) The illustration of FIG. **19** shows a sectional view of a lid **90** according to the invention and according to a sixth embodiment of the invention from obliquely below. The lid **90** is of a configuration which is substantially identical to that of the lid **20** of FIGS. **7** to **9**. Therefore, only the points of differentiation in comparison to the lid **20** of FIGS. **7** to **9** will be explained.

(61) As opposed to the lid **20** of FIGS. **7** to **9**, the periphery **92** of the second ply **26** of the lid collar **14** is not configured so as to be smooth but is provided with a wave cut, in other words configured so as to be wave-shaped, see also FIG. **20**. In other words, the periphery **92** of the second ply **26** of the lid collar **14** has a sinusoidal profile. As a result, portions of the periphery **92** come to engage with the groove **28** once the second ply **26** has been folded over by 180° at the location of the periphery **24** of the lid collar **14**. However, the engagement of only portions of the periphery **92** in the groove **28** is sufficient in order to achieve secure fixing of the second ply **26** in the position illustrated in FIG. **20**. The wave shape, or the wave cut, of the periphery **92** minimizes material stresses when folding over the second ply **26**. As a result, deformations during folding over are reduced, and the folding over of the second ply **26** is facilitated.

(62) The portions of the periphery **92** that are disposed in the groove **28** can additionally be secured in the groove **28** by an adhesive. The two plies **22**, **26** of the lid collar **14** can also be fixed to each other by sealing when the planar blank from which the lid **90** is produced is at least partially coated with a sealable plastics material or a sealable dispersion coating. In principle however, the lid **90** can also be produced without the utilization of an adhesive, and also with a non-coated blank from paper, paperboard or paper-like material, or with a blank from paper, paperboard or paper-like material that is only coated with a protective varnish against invading liquids. This is because the latching of portions of the periphery **92** in the groove **28** ensures secure mechanical fixing of the second ply **26** and a stable embodiment of the lid collar **14** and of the lid **90** overall.

(63) FIG. 21 shows a lid **100** according to the invention and according to a further embodiment. The lid **100** is configured as a so-called snap-in lid and is inserted into the open end of a sleeve or of a container such that lower portion **102** of a lid collar in FIG. 21 comes to lie on the inside of the external wall of the container. An upper portion **104** of the lid collar is widened in a radially outward manner. As a result, an encircling shoulder is configured at the transition between the lower portion **102** and the upper portion **104**, said shoulder forming a detent which restricts the depression depth of the lid **100** when the lid **100** is pressed into the open end of a container or of a sleeve. The lid **100** is integrally configured and has a lid plate **106** and the lid collar **108** which protrudes upward from the lid plate **106** so as to be substantially perpendicular to the latter.

(64) The lid collar **108** is configured so as to have two plies in the lower portion **102** as well as in the upper portion **104**, see also FIG. 23. The lid collar **108** is folded over by 180° on the upper periphery thereof in FIG. 21, and the folded-over portion is then latched into a groove on the periphery of the lid plate **106**, said groove not being visible in FIG. 21. The deformation of the two-ply lid collar **108** can take place simultaneously or subsequently such that the lower portion **102** and the upper portion **104**, widened in a radially outward manner, are formed.

(65) An encircling shoulder **110** is likewise configured on the inside of the lid **100**. This shoulder **110** serves as a detent when stacking a plurality of lids **100** of the same type.

(66) FIG. 22 shows a total of four lids **100** of the same type that are nested in one another. It can be seen that the respective upper lid **100** can be pushed into a lower lid **100** only so far until the lower periphery of the lid collar in FIG. 21, which then transitions to the lid plate **106**, bears on the encircling inner shoulder **110**.

(67) This can also be seen in FIG. 23. In this way, numerous lids **100** can be stacked without the risk of the stacked lids **100** being wedged in one another. This is a substantial advantage in particular when the lids **100** are to be removed rapidly and without problems from a high storage stack so as to be placed onto a container filled with a beverage. As a result, the lids **100** can be removed by hand from a very high stack with numerous lids **100**.

(68) FIG. 24 shows a further lid **120** according to the invention and according to a further embodiment of the invention. The lid **120** is likewise configured as a snap-in lid, such that in the state placed on a container a portion **122** of the lid collar by way of the external side thereof bears on an inside of the shell of a container or of a sleeve. The lid collar **122** proceeds from a lid plate **124**. The lid **120** is integrally configured, see also FIGS. 25 and 26.

(69) A groove **126** in which the peripheral end portion **128** of the lid collar is received is provided on the outer periphery of the lid plate **124**. FIG. 26 shows the detail A of the sectional view of the lid **120** from FIG. 24, and it can be seen that the peripheral end portion **128** of the lid collar is latched in the groove **126**. The lid collar **122** has a first, outer ply **130** which proceeds from the lid plate, or the groove **126** in the lid plate. The outer ply **130** is angled by 90° in relation to the lid plate **124**. The lid collar **122** has a second, inner ply **132** which is disposed parallel to the outer ply **130** and which on the upper periphery of the lid collar **122** is connected to the outer ply **130**. The peripheral end portion **128** forms the lower peripheral end portion of the inner ply **132**. As a result of the peripheral end portion **128** being received in the groove **126** and latched in the latter, the inner ply **132** cannot move away from the position illustrated in FIG. 26. As a result, the lid **120** can be produced from a non-coated paper material because the inner ply **122** of the lid collar **122** is mechanically held so as to be parallel to the outer ply **130**.

(70) The lid collar **122** on the periphery thereof that is spaced apart from the lid plate **124** and lies at the top in FIG. 26 has a protrusion **134** that extends radially outward. This protrusion **134** forms a detent which delimits the depression depth of the lid **120** when the latter is pressed into the open end of a sleeve or of a container.

(71) The lid collar **122** on the upper end thereof in FIG. 26 also has a protrusion **136** that protrudes radially inward. This radially inward protruding protrusion **136** forms an extraction aid. In the depressed state of the lid **120**, a user can grip below the protrusion **136** with one or a plurality of

fingers and, as a result, extract the lid **120** in a simple manner upward from the container or from the sleeve.

(72) In order for the radially outward extending protrusion **134** to be shaped, the outer ply of the lid collar **130** is bent radially outward by 90° and after a short distance of a few millimeters bent inward by 180° again. For shaping the radially inward extending protrusion **136**, the inner ply **132** is bent downward by 90°, then guided at an angle of almost 45° in relation to the outer ply **130** and finally bent again such that said inner ply **132** runs parallel to the outer ply **130** and into the groove **126**. As a result, the radially inward extending protrusion **136** has a significantly larger bending radius than the radially outward extending protrusion **134** and, as a result, can be more easily gripped by a human finger. In the context of the invention however, it is of course also possible for the radially outward extending protrusion **134** and the radially inward protruding protrusion **136** to be configured in the same way and to be mutually symmetrical, for example.

(73) FIG. **27** shows a lid **140** according to the invention and according to a further embodiment. The lid **140** is configured as a snap-in lid and likewise produced from an integral, planar blank. A lid collar **142** in the region of the periphery thereof that is spaced apart from the lid plate **144** has two radially outward extending protrusions **146**, **148**. The protrusions **146**, **148** are configured by bending the outer ply of the lid collar **142** radially outward by 90° and then, following a short distance of a few millimeters, again radially inward by 180°, see FIG. **31**. In the region of the radially outward extending protrusions **146**, **148**, the inner ply of the lid collar is then bent downward again by 90° and then runs parallel to the outer ply of the lid collar up to the groove **150** on the outer periphery of the lid plate **144**. The peripheral end portion of the inner ply of the lid collar **142** is received in the groove **150**, see FIG. **30** and FIG. **31**.

(74) The radially outward extending protrusions **146**, **148** are disposed so as to be mutually opposite and extend in each case only across an angular range of approximately 130° about the circumference of the lid **140**, see FIG. **29**. Two mutually opposite and radially inward extending protrusions **156**, **158** are disposed between the two radially outward extending protrusions **146**, **148**. The protrusions **156**, **158** form in each case an extraction aid for a user. A user can grip below one of the protrusions **156**, **158** with one or a plurality of fingers and, as a result, lift the lid **140** from a container.

(75) FIG. **28** shows the lid **140** in a state in which the latter closes a container **160**. The container **160** is configured as a cylindrical sleeve. It can be seen that the lid **140** is depressed into the container **160** up the radially outward extending protrusions **146**, **148**. The lower sides of the protrusions **146**, **148** in this instance form a detent such that the lid **140** cannot be pressed farther into the container **160**. As a result, an external face of the lid collar **142** in portions bears on the internal face of a shell of the container **160**.

(76) As has already been discussed, the lid **140** can be extracted from the container **160** in that one reaches below one of the two protrusions **156**, **158** and the lid **140** in FIG. **28** is extracted upward from the container **160**.

(77) FIG. **32** shows portions of a sectional view of the assembly of FIG. **28**. The container **160** can be seen in the region of the upper open end thereof. The section plane in FIG. **32** runs through the radially inward extending protrusion **158**. Portions of the radially outward extending protrusion **148** can likewise be seen.

(78) The radially inward extending protrusion **158** is configured in that the outer ply of the lid collar **142** is bent radially inward by 90° and after a horizontal profile of a few millimeters, for example 5 mm is folded over in a radially outward manner by 180° again. The protrusion **158** is configured in a two-ply manner as a result. The inner ply of the lid collar **142** is then disposed so as to be parallel to the outer ply again, and the periphery of the inner ply is received in the groove which is disposed on the outer periphery of the lid plate **144**.

(79) FIG. **33** shows a section through the assembly of FIG. **28**, wherein the section plane runs through the radially outward extending protrusion **148**. It can be seen that the external face of the

lid collar **142** bears in a planar manner on the internal face of the external wall of the container **160**. It can also be seen that the lid **140** is depressed into the container **160** so far that the lower side of the radially outward extending protrusion **148** lies against the upper periphery of the external shell of the container **160**.

(80) FIG. **34** shows a lid **170** according to a further embodiment of the invention. The lid **170** is configured as a snap-in lid and produced from an integral, planar blank. The lid **170** in terms of construction is very similar to the lid **140** which has already been described by means of FIGS. **27** to **33**. Therefore, only the points of differentiation in comparison to the lid **140** will be explained hereunder.

(81) FIG. **35** shows an assembly with a container **160** which is configured as a cylindrical sleeve having a circular cross section. The lid **170** is depressed into the upper open end of the container **160**.

(82) The lid **170** has a lid collar **172**, wherein the lid collar **172** has a lower portion **174**, the external circumference of the latter being smaller than the internal circumference of the container **160** or being exactly the same size as the internal circumference of the container **160**. The lid collar **172** has an upper portion which has been deformed in a radially outward manner and, as a result, has two mutually opposite protrusions **176**, **178**. At the transition from the lower portion **174** of the lid collar **172** to the upper portion, which is widened in a radially outward manner, the lower side of the protrusion **176** forms a shoulder, or a detent, respectively, wherein this shoulder delimits the depression depth of the lid **170** into the container **160**. This radially outward extending shoulder **180** encircles the entire circumference of the lid collar **172**.

(83) Two radially inward extending protrusions **182**, **184** are disposed so as to be mutually opposite between the radially outward extending protrusions **176**, **178**. The radially inward extending protrusions **182**, **184** serve as an extraction aid. A user can grip below one or both of the protrusions **182**, **184**, see FIG. **35**, and as a result extract the lid **170** upward from the container **160**.

(84) FIG. **36** shows portions of a sectional view of the assembly of FIG. **35**, wherein a section plane runs through the radially outward extending protrusion **178**. It can be seen that the shoulder, which is located at the transition between the lower portion of the lid collar **172** and the upper portion of the lid collar **172** and extends radially outward, forms a detent up to which the lid **170** can be depressed into the open end of the container **160**.

(85) FIG. **37** shows portions of a sectional view of the assembly of FIG. **35**, wherein the section plane runs through the radially inward extending protrusion **180**. Portions of the radially outward extending protrusion **178** can likewise be seen.

(86) As has been explained, the shoulder between the lower portion of the lid collar **172** and the upper portion of the lid collar **172**, which has been deformed in a radially outward manner, encircles the entire circumference of the lid **170**. As a result, the lid **170** sits reliably on the container **160** and as a result can also reliably seal the latter.

(87) It can be seen in FIG. **36** as well as FIG. **37** that a lid plate **188** on the outer periphery thereof has a groove **190** in which the peripheral end portion of the inner ply of the lid collar **172** is received and, as a result, mechanically latched

(88) As a result, the lid **170**, like the lids **100**, **120**, **140** already described, can be configured from non-coated paper material.

(89) In the production of the lids **10**, **50**, **60**, **40**, **80**, **90**, **100**, **120**, **140**, **170** according to the invention, an integral, planar circular blank is initially deformed to a pot-type intermediate element. In the production of a circular lid as is illustrated in FIGS. **1** to **37**, the planar blank is circular. In the production of a lid with a different shape, for example rectangular or oval, the planar blank of course has a different shape. After the pot-type intermediate element is configured, the encircling wall assembly of the pot-type element is folded, or folded over, such that a first ply of the lid collar, which in the radial direction is on the outside, and a second ply of the lid collar, which in the radial

direction is on the inside, are formed. The wall assembly of the pot-type element here can be folded over by 180°, but in the context of the invention does not necessarily have to be folded over by 180°. Angles of less than 180° are possible.

(90) Before or after the encircling wall assembly of the pot-type element has been folded over, or else even before the pot-type element is configured, a groove is formed in a radially outer region of the base of the pot-type intermediate element. This can be performed by embossing an encircling groove in the still flat, planar blank, for example. A peripheral end portion of the second ply of the lid collar is then latched into this groove once the encircling wall assembly of the pot-type element has been folded over. As a result the second ply is fixed in the position thereof, even without the utilization of an adhesive.

(91) The groove can be molded in such a way that a portion of the first ply of the lid collar forms a lateral wall of the groove. As a result, the groove is disposed directly on the periphery of the lid plate.

(92) In the context of the invention, the groove may also be configured somewhat within the outer periphery of the lid plate. In this instance, the second ply of the lid collar is not folded over by 180° but by a smaller angle. The groove does also not necessarily have to be embossed or molded from the lower side of the lid plate. For example, the groove can also be formed between a protrusion that protrudes beyond the lower side of the lid plate **12** and the inside of the first ply of the lid collar.

(93) In the method according to the invention, the groove can be shaped in such a way that an encircling protrusion is configured on an upper side of the lid plate that lies opposite the lid collar, wherein the groove is disposed on a side of the protrusion that faces the lower side of the lid plate. This protrusion can form the outer periphery of the lid plate and, as a result, simultaneously contributes toward reinforcing the lid plate.

(94) The peripheral end portion of the second ply, at least in portions, can be fixed in the groove by means of an adhesive. To this end, an adhesive bead from hot-melt adhesive can be incorporated in the groove before the peripheral end portion of the second ply latches in the latter, for example.

(95) In the production of the lid according to the invention, the lid plate can be shaped by means of embossing the base of the pot-type intermediate element. In the process, embossings or stampings can be performed for reinforcing the lid plate, for example, or also for attaching a drinking opening, for example, or only attaching a perforation in the region of which a drinking opening can then be placed by a user. On the lid collar, protrusions which extend radially outward or which extend radially inward can be formed.

Claims

1. A lid from paper, paperboard or paper-like material for a container, wherein the lid is formed from a one-piece integral planar blank and has a lid plate and a lid collar, the lid collar projecting from a periphery of the lid plate, the lid collar being formed from at least two plies of the one-piece integral planar blank, a first ply of the lid collar, starting from the periphery of the lid plate, extending to a peripheral edge portion of the lid collar, a second ply of the lid collar being disposed on an inner side of the lid collar, the second ply starting from the peripheral edge portion of the lid collar and extending towards the lid plate and terminating at a second ply peripheral end portion, the second ply peripheral end portion having at least a portion latched in a groove by a latching action, the groove being disposed in a region of the periphery of the lid plate and being defined solely by the one-piece integral planar blank, the at least a portion of the second ply peripheral end portion being fixed in the groove solely by the latching action, and further wherein: the second ply peripheral end portion is configured with clearances or is configured so as to be wave-shaped; and/or the second ply has incisions, the incisions starting from the second ply peripheral end

portion, and portions of the second ply disposed between the incisions are configured so as to extend obliquely radially inwardly.
